Class 2: Basics of Graph Plotting

Summer Break Bridge Course

Why Graphs Matter

In the previous class we discussed different forms of scientific representation: words, numbers, tables, equations, and graphs. A graph is powerful because it allows us to **see the behavior of a mathematical rule**. It helps us answer questions such as:

- Is the function increasing or decreasing?
- Is it bounded or unbounded?
- Is it symmetric?
- Does it oscillate?
- Does it blow up or flatten out?
- What happens at large positive or negative values of x?

A graph is not a picture to memorize. It is the visible behavior of a rule.

Independent and Dependent Variables

When we draw a graph of

$$y = f(x),$$

we usually think of:

- x as the **independent variable**,
- y or $f(x)$ as the **dependent variable**.

The horizontal axis is usually the x-axis, and the vertical axis is the y-axis.

Sign Information from the Four Quadrants

A rough graph already tells us a lot:

- First quadrant: $x > 0,\ y > 0$
- Second quadrant: $x < 0,\ y > 0$
- Third quadrant: $x < 0,\ y < 0$
- Fourth quadrant: $x > 0,\ y < 0$

So while sketching, one of the first things to check is: *for which x-values is the function positive, negative, or zero?*

Some Basic Functions and Their Shapes

We begin with a few standard functions. These are not merely to be memorized. They are the basic building blocks from which we understand more complicated graphs.

1. Linear Function

$$f(x) = x$$

This is a straight line passing through the origin with slope 1.

2. Quadratic Function

$$f(x) = x^2$$

This is a parabola opening upward. It is always nonnegative and symmetric about the y-axis.

3. Cubic Function

$$f(x) = x^3$$

This passes through the origin, is odd, and changes sign across $x = 0$. For large positive x, it grows to $+\infty$; for large negative x, it goes to $-\infty$.

4. Square-root Function

$$f(x) = \sqrt{x}$$

This exists only for $x \geq 0$. It increases but slowly bends over as x becomes larger.

5. Constant Function

$$f(x) = a$$

Its graph is a horizontal line.

6. Reciprocal Function

$$f(x) = \frac{1}{x}$$

This is a hyperbola. It is undefined at $x = 0$, and it approaches the coordinate axes without touching them.

7. Modulus Function

$$f(x) = |x|$$

This is a V-shaped graph. It can be thought of as

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

8. Exponential Growth and Decay

$$f(x) = e^x, \qquad f(x) = e^{-x}$$

The function e^x grows rapidly as $x \to \infty$, while e^{-x} decays to zero as $x \to \infty$.

9. Logarithm

$$f(x) = \ln x$$

This exists only for $x > 0$. It increases slowly and goes to $-\infty$ as $x \to 0^+$.

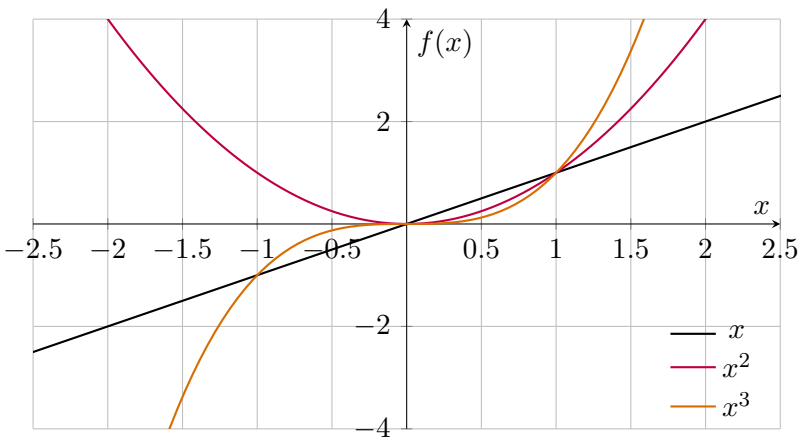
10. Sine Function

$$f(x) = \sin x$$

This is periodic and oscillatory, taking values between -1 and 1.

Example Plots of the Standard Functions

Linear, Quadratic, and Cubic

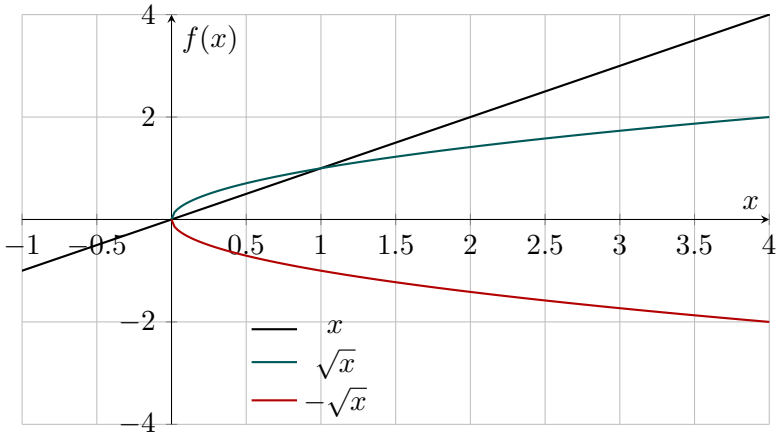


These three are good to compare together:

- x changes at a constant rate,

- x^2 bends upward and stays nonnegative,
- x^3 behaves somewhat like x near the origin but grows much faster for large $|x|$.

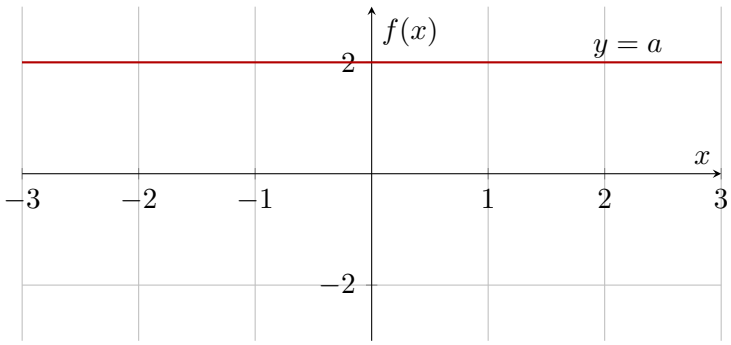
Square-root Function and Comparison with x



A useful comparison:

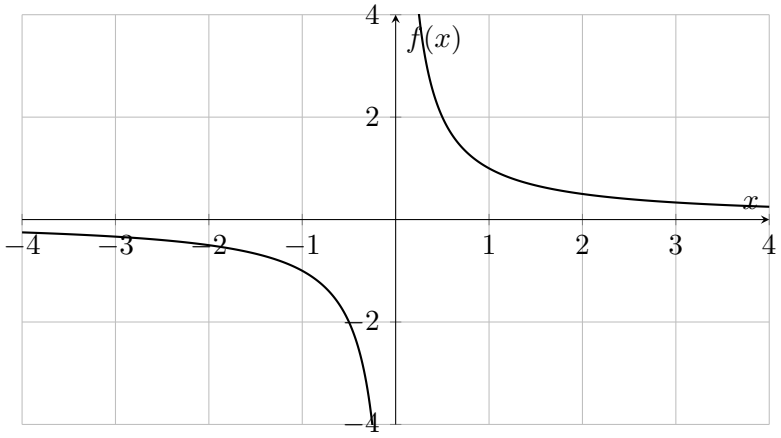
$$\sqrt{x} > x \quad \text{for} \quad 0 < x < 1, \qquad \sqrt{x} < x \quad \text{for} \quad x > 1.$$

Constant Function



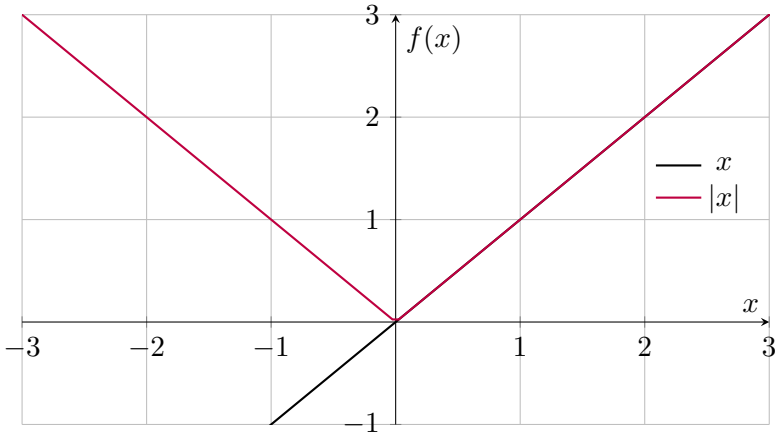
A constant function does not change with x. Its graph is horizontal.

Reciprocal Function



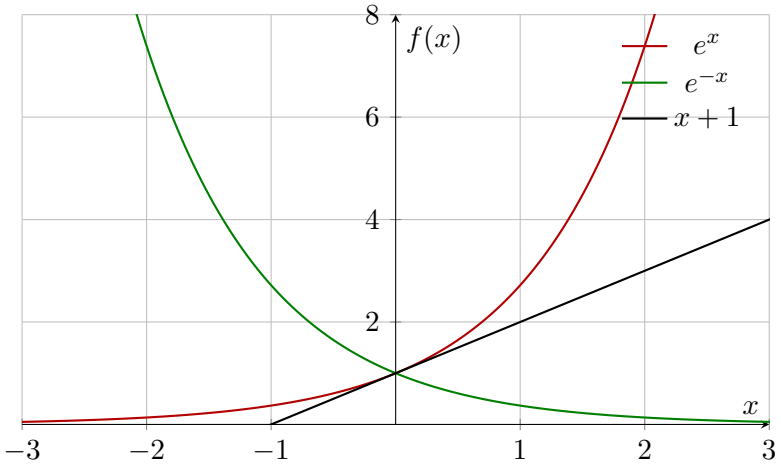
This function is undefined at $x = 0$, and its graph approaches both axes without touching them.

Modulus Function and Comparison with x



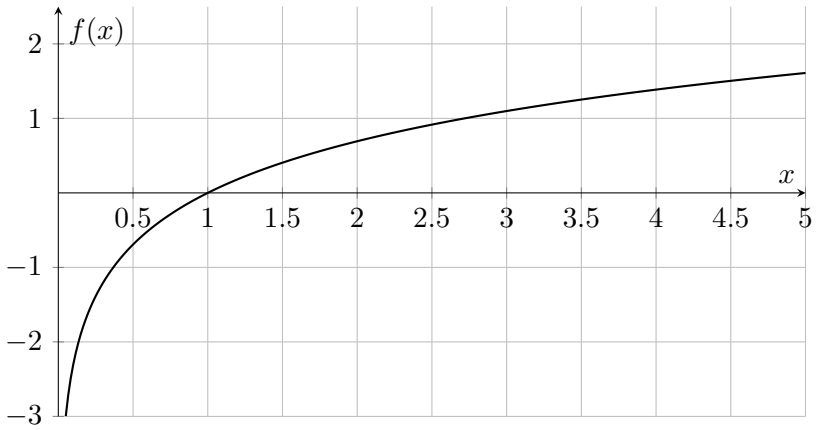
The negative part of x gets folded upward to produce $|x|$.

Exponential Growth and Decay



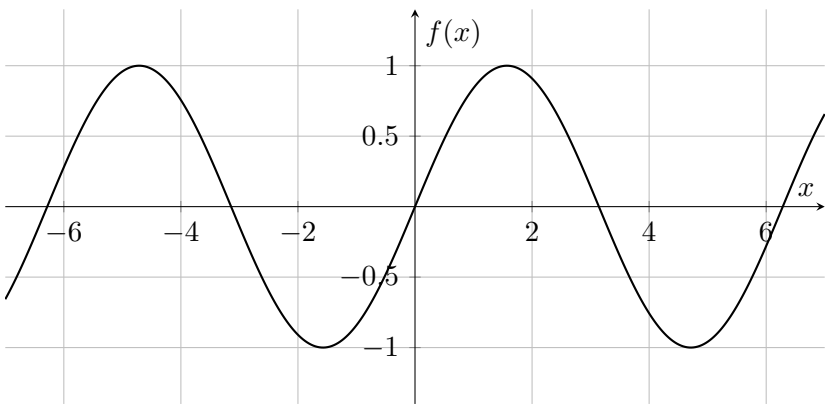
Here e^x shows rapid growth, while e^{-x} shows decay.

Logarithm



The logarithm exists only for $x > 0$. It grows slowly and drops to $-\infty$ as $x \to 0^+$.

Sine Function



This is a periodic oscillatory function:

$$-1 \leq \sin x \leq 1.$$

Domain and Range

Domain

The **domain** of a function is the set of allowed input values.

Range

The **range** of a function is the set of possible output values.

Function	Domain	Range
x	$(-\infty, \infty)$	$(-\infty, \infty)$
x^2	$(-\infty, \infty)$	$[0, \infty)$
$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$\ln x$	$(0, \infty)$	$(-\infty, \infty)$
$\sqrt{-x}$	$(-\infty, 0]$	$[0, \infty)$
$\sqrt{\lvert x \rvert}$	$(-\infty, \infty)$	$[0, \infty)$
$\sin x$	$(-\infty, \infty)$	$[-1, 1]$
$\frac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$

Whenever you see a square root, a logarithm, or a denominator, always ask first: **what values of x are allowed?**

Symmetry of Functions

Symmetry is one of the fastest ways to understand a graph.

Even Functions

A function is **even** if

$$f(-x) = f(x).$$

Its graph is symmetric about the y-axis.

Examples:

$$x^2, \qquad |x|, \qquad \cos x.$$

Odd Functions

A function is **odd** if

$$f(-x) = -f(x).$$

Its graph is symmetric about the origin.

Examples:

$$x, \qquad x^3, \qquad \sin x.$$

For an even function, once you know the graph on one side of the y-axis, the other side is just its mirror image.
For an odd function, once you know the graph on one side, the other side is obtained by reflection through the origin.

Not Every Function is Even or Odd

Many functions are neither. However, any function $f(x)$ can be written as a sum of an even part and an odd part:

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}.$$

Define

$$g(x) = \frac{f(x) + f(-x)}{2}, \qquad h(x) = \frac{f(x) - f(-x)}{2}.$$

Then

$$f(x) = g(x) + h(x),$$

where $g(x)$ is even and $h(x)$ is odd.

Bounded and Unbounded Functions

A function is **bounded** if its values stay within some upper and lower limits.

For example:

$$-1 \leq \sin x \leq 1,$$

so $\sin x$ is bounded.

But x^2 is unbounded above, and x is unbounded both above and below.

Monotonic and Non-monotonic Behavior

Monotonically Increasing

A function is monotonically increasing if larger x always gives larger $f(x)$.

Example:

$$f(x) = x, \qquad f(x) = e^x.$$

Monotonically Decreasing

A function is monotonically decreasing if larger x always gives smaller $f(x)$.

Example:

$$f(x) = e^{-x}.$$

Non-monotonic

A function is non-monotonic if it does not keep increasing all the time or decreasing all the time.

Examples:

$$x^2, \qquad \sin x, \qquad x^3 - 3x.$$

Special Non-monotonicity: Periodic/Oscillatory

Some functions show a repeating pattern of increase and decrease. The sine function is a classic example.

Large-x Behavior

When sketching an unknown function, it is very useful to ask:

- What happens as $x \to +\infty$?
- What happens as $x \to -\infty$?

This is often called the **large-x behavior** or **end behavior** of a function.

Examples:

$$x^2 \to \infty \quad \text{as} \quad x \to \pm\infty,$$
$$\frac{1}{x} \to 0 \quad \text{as} \quad x \to \pm\infty,$$
$$e^{-x} \to 0 \quad \text{as} \quad x \to +\infty.$$

Anchor Points and Special Points

When you do not know the full graph, it is smart to begin with a few anchor points.

Useful Anchor Questions

- What is $f(0)$? (the y-intercept)
- What is $f(1)$?
- Where does $f(x) = 0$? (the x-intercepts)

Special Points

Also watch for special points such as:

- divergence points, for example $x = 1$ in $\dfrac{1}{x-1}$,
- corners, for example $x = 0$ in $|x|$,
- points where domain begins or ends, for example $x = 0$ in $\sqrt{x}$ or $\ln x$.

A Practical Checklist for Sketching an Unknown Function

Whenever you see a function, do not panic. Ask:

1. What is the domain?
2. Is it even, odd, or neither?
3. Is it positive, negative, or zero in different regions?
4. What are a few anchor points?
5. Is it increasing, decreasing, oscillatory, or mixed?
6. What happens for large $|x|$?
7. Are there any divergences, corners, or domain boundaries?

A rough sketch built from these clues is often far more useful than trying to memorize graph shapes blindly.

Transformations of Graphs

This part is extremely important. If we know the graph of $y = f(x)$, then we can quickly predict the graph of many related functions without re-plotting from scratch.

1. Adding a Constant to the Function: Vertical Shift

$$y = f(x) + c$$

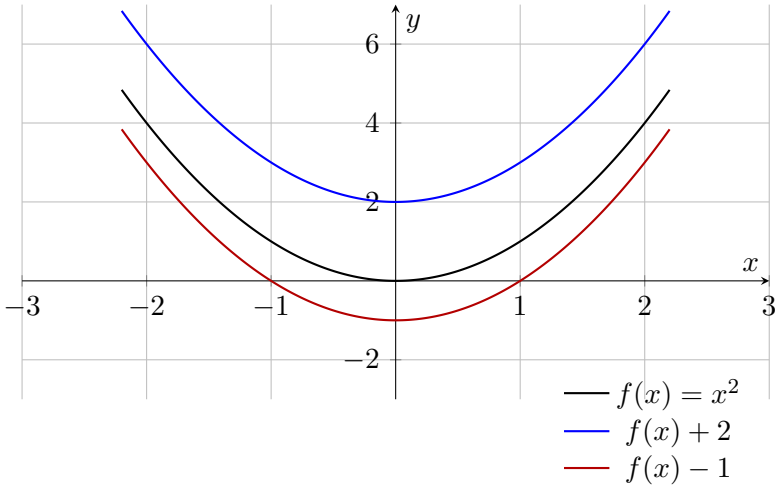
This shifts the graph vertically.

- If $c > 0$, the graph shifts **up** by c units.
- If $c < 0$, the graph shifts **down** by $|c|$ units.

If $f(x) = x^2$, then

$$y = x^2 + 3$$

is the parabola shifted upward by 3 units.



2. Adding a Constant to the Input: Horizontal Shift

$$y = f(x + c)$$

This shifts the graph horizontally, but the direction is the opposite of what many students first guess:

- If $c > 0$, the graph shifts **left** by c units.
- If $c < 0$, the graph shifts **right** by $|c|$ units.

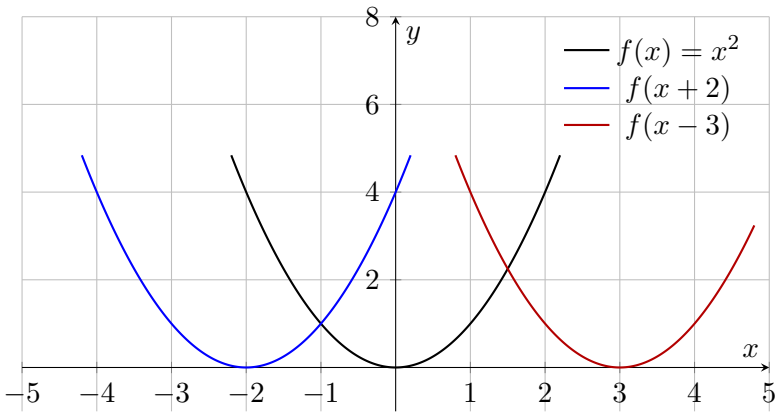
If $f(x) = x^2$, then

$$y = (x + 2)^2$$

is the parabola shifted left by 2 units, while

$$y = (x - 3)^2$$

is the parabola shifted right by 3 units.



Why is the direction opposite?

Suppose the original graph $y = f(x)$ has a special point at $x = a$. In the new graph

$$y = f(x + c),$$

the same output is obtained when

$$x + c = a \quad \Rightarrow \quad x = a - c.$$

So every feature moves from $x = a$ to $x = a - c$. Hence $+c$ inside the function means a shift to the left.

3. Multiplying the Output by a Constant: Vertical Stretch/Compression

$$y = c\,f(x)$$

- If $|c| > 1$, the graph stretches vertically.
- If $0 < |c| < 1$, the graph compresses vertically.
- If $c < 0$, there is also a reflection about the x-axis.

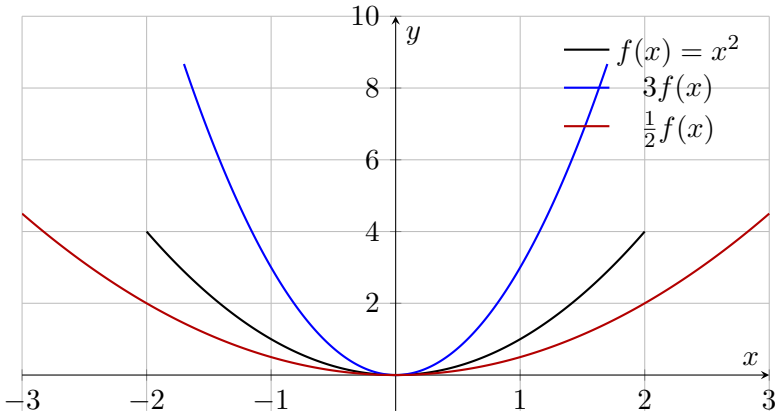
If $f(x) = x^2$, then

$$y = 3x^2$$

is a vertically stretched parabola, while

$$y = \frac{1}{2}x^2$$

is vertically compressed.



4. Multiplying the Input by a Constant: Horizontal Stretch/Compression

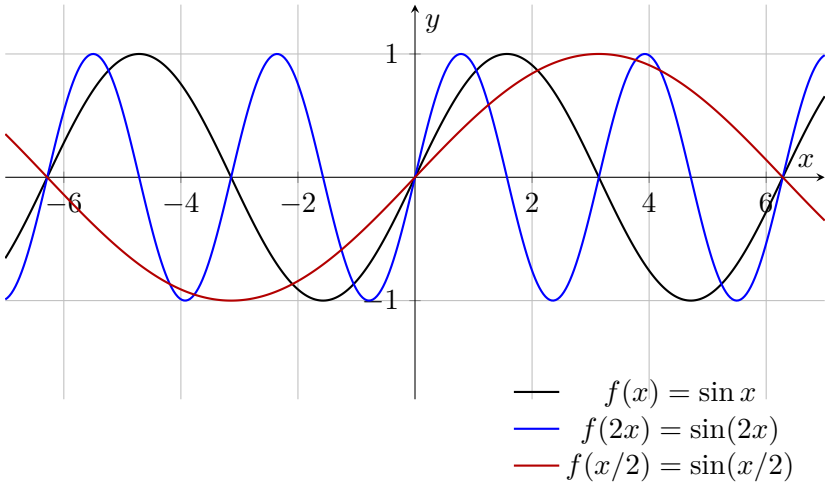
$$y = f(cx)$$

- If $|c| > 1$, the graph compresses horizontally.
- If $0 < |c| < 1$, the graph stretches horizontally.
- If $c < 0$, there is also a reflection about the y-axis.

If $f(x) = \sin x$, then

$$y = \sin(2x)$$

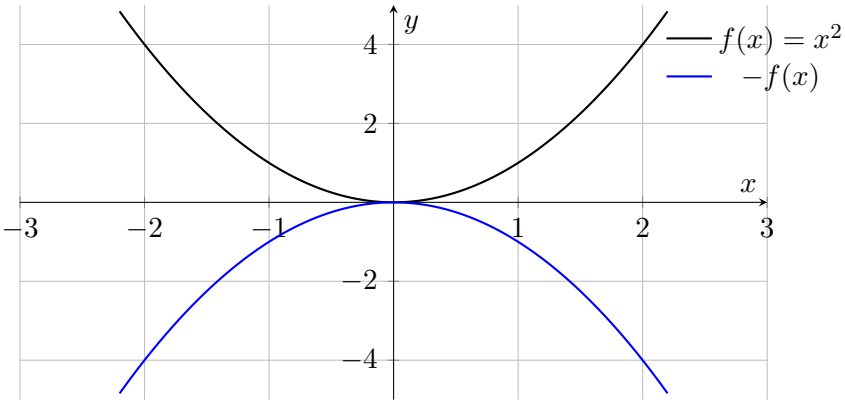
oscillates twice as fast, so its graph is horizontally compressed.



5. Reflection About the x-Axis

$$y = -f(x)$$

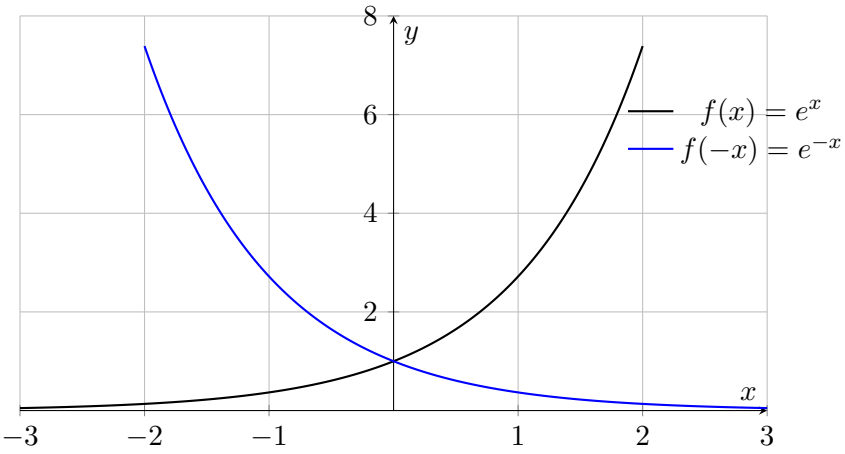
This flips the graph vertically.



6. Reflection About the y-Axis

$$y = f(-x)$$

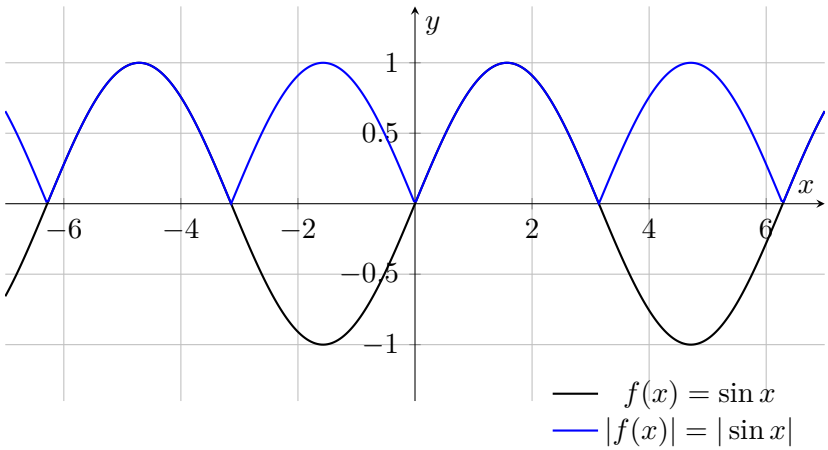
This flips the graph horizontally.



7. Absolute Value of the Output

$$y = |f(x)|$$

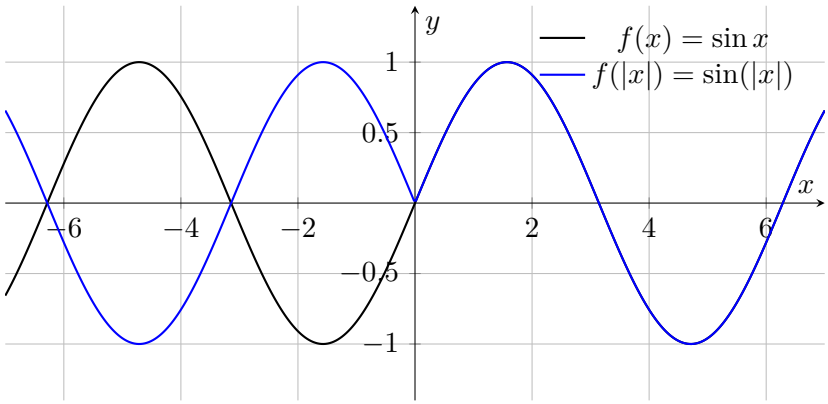
Wherever the original graph lies below the x-axis, that part is folded upward.



8. Absolute Value of the Input

$$y = f(|x|)$$

The right half of the graph is mirrored onto the left. This always produces an even function.



A useful rule of thumb:

- changes **outside** $f(\cdot)$ affect the graph in the usual direction,
- changes **inside** $f(\cdot)$ affect the graph in the opposite horizontal direction.

A Short Transformation Summary Table

New function	Effect on graph
$f(x) + c$	Shift up/down by c
$f(x + c)$	Shift left/right by $-c$
$c\,f(x)$	Vertical stretch/compression; reflect if $c < 0$
$f(cx)$	Horizontal compression/stretch; reflect if $c < 0$
$-f(x)$	Reflection about x-axis
$f(-x)$	Reflection about y-axis
$\lvert f(x)\rvert$	Fold negative y-part upward
$f(\lvert x\rvert)$	Mirror right half to the left

Basic Graph Plotting in Python

In class we also introduced very basic function plotting in Python. The purpose here is not advanced programming. It is to connect mathematical functions with actual plotted output.

Main Libraries

Two important libraries are:

- `numpy` — for arrays and numerical data,
- `matplotlib.pyplot` — for plotting.

Step 1: Import

```
import numpy as np
import matplotlib.pyplot as plt
```

Step 2: Create x-values

There are two common ways.

Manual array

```
x_array = np.array([0, 1, 2, 3, 4, 7, 9.5])
```

Evenly spaced points in an interval

```
npoints = 500
x_low = -5
x_high = 5
x_array = np.linspace(x_low, x_high, npoints)
```

Here `np.linspace(a,b,n)` creates n equally spaced points from a to b.

Fixed step size

```
dx = 0.5
x_array = np.arange(-5, 5, dx)
```

Here `np.arange(a,b,dx)` creates points with spacing dx.

Step 3: Define a Function

```
def linear_function(x):
    return 2*x
```

Or for another example:

```
def f1(x):
    return x**3 - 5*x**2 + 2*x + 4
```

Step 4: Compute y-values

```
y_array = linear_function(x_array)
```

Step 5: Plot

```
plt.plot(x_array, y_array, lw=3)
plt.xlabel("x")
plt.ylabel("y")
plt.grid()
plt.show()
```

Changing Appearance

You can change line width, color, and line style:

```
plt.plot(x_array, y_array, lw=4, color='red', ls='--')
plt.xlabel("x", size=16)
plt.ylabel("y", size=16)
plt.xlim(-3, 3)
plt.ylim(-6, 6)
plt.grid()
plt.show()
```

Plotting Multiple Functions on the Same Graph

```
def f1(x):
    return np.exp(x)

def f2(x):
    return x**2

def f3(x):
    return np.sin(x)

x_array = np.linspace(-3, 3, 500)

y1 = f1(x_array)
y2 = f2(x_array)
y3 = f3(x_array)

plt.plot(x_array, y1, lw=3, color='black', label='Exponential')
plt.plot(x_array, y2, lw=3, color='red', label='Quadratic')
plt.plot(x_array, y3, lw=3, color='blue', label='Sinusoidal')

plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.grid()
plt.show()
```

Python plotting follows a simple pattern:

$$\text{make } x\text{-data} \to \text{compute } y\text{-data} \to \text{plot}$$

Final Perspective

Graph plotting is not only about drawing known shapes like a line or a parabola. It is about learning how to read a function.

A useful graphing mindset is:

1. understand the domain,
2. locate anchor points,
3. check symmetry,
4. look at sign and monotonicity,

5. inspect large-x behavior,
6. identify transformations from familiar graphs,
7. then use Python or rough sketching to visualize the result.

Key Takeaway:
A graph is a way of thinking. Once you learn to read how a function behaves, plotting becomes much more intuitive than memorizing isolated shapes.